

```
# =====  
# Facial Recognition Pipeline using Computer Vision Levels  
# Google Colab  
# =====  
  
import cv2  
import matplotlib.pyplot as plt  
from google.colab import files  
  
# Upload image  
uploaded = files.upload()  
  
image_name = list(uploaded.keys())[0]  
  
# Read image  
img = cv2.imread(image_name)  
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)  
  
# -----  
# Low-Level Vision: Convert to Grayscale  
# -----  
  
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)  
  
# -----  
# Mid-Level Vision: Face Detection  
# -----  
  
face_cascade = cv2.CascadeClassifier(  
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"  
)  
  
faces = face_cascade.detectMultiScale(  
    gray,  
    scaleFactor=1.1,  
    minNeighbors=5,  
    minSize=(30,30)  
)  
  
# Draw rectangles around detected faces  
result = img_rgb.copy()  
  
for (x, y, w, h) in faces:  
    cv2.rectangle(result,  
        (x, y),  
        (x+w, y+h),  
        (0,255,0),  
        2)
```

```
# -----  
# Display Images  
# -----  
  
plt.figure(figsize=(12,5))  
  
plt.subplot(1,2,1)  
plt.imshow(img_rgb)  
plt.title("Original Image")  
plt.axis("off")  
  
plt.subplot(1,2,2)  
plt.imshow(result)  
plt.title("Detected Face(s)")  
plt.axis("off")  
  
plt.tight_layout()  
plt.show()  
  
# -----  
# Output  
# -----  
  
print("Facial Recognition Pipeline")  
print("-----")  
print("1. Low-Level Vision")  
print("  - Image captured by camera.")  
print("  - Converted to grayscale for preprocessing.")  
  
print("\n2. Mid-Level Vision")  
print("  - Face features extracted.")  
print("  - Face detected using Haar Cascade.")  
  
print("\n3. High-Level Vision")  
print("  - Face is recognized by comparing")  
print("    extracted features with a database.")  
print("  - Decision: Identify or Verify the person.")  
  
print(f"\nNumber of Faces Detected: {len(faces)}")
```

Choose Files number_plate.jpg

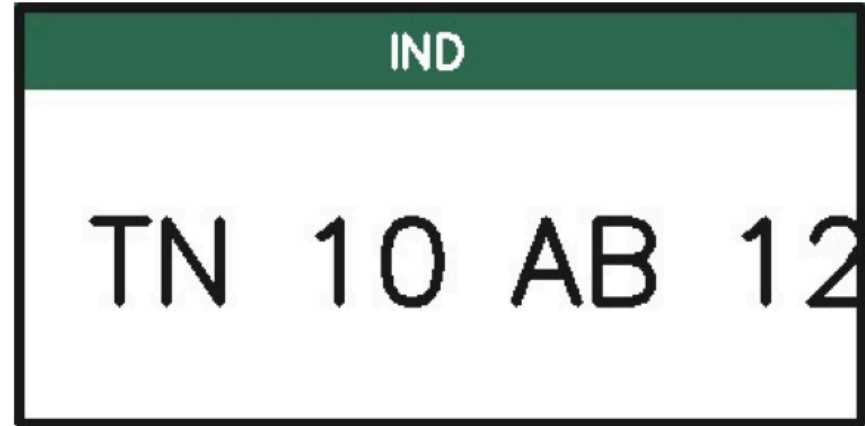
number_plate.jpg(image/jpeg) - 14891 bytes, last modified: 7/13/2026 - 100% done

Saving number_plate.jpg to number_plate.jpg

Original Image



Detected Face(s)



Facial Recognition Pipeline

1. Low-Level Vision

- Image captured by camera.
- Converted to grayscale for preprocessing.

2. Mid-Level Vision

- Face features extracted.
- Face detected using Haar Cascade.

3. High-Level Vision

- Face is recognized by comparing extracted features with a database.
- Decision: Identify or Verify the person.

Number of Faces Detected: 0